



Zewail City of Science and Technology

REE 419 Power Systems

Power System Analysis Tool

Load Flow and Short Circuit Study

Student Names: Walid Sherif 202200702

Instructor: Mohamed Morgan

TA: Sara Badreldin

Date: August 25, 2026

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1. Introduction

This project studies a five-bus power system using load flow and short-circuit analysis. The main objective is to calculate the voltage magnitude and angle at each bus, the active and reactive power generated by the slack and PV buses, the current flow in each branch, and the total system active and reactive power losses. The Newton–Raphson method was implemented in MATLAB, while PowerWorld Simulator was used to validate the results.

The project is divided into five main tasks. First, the base-case load flow is solved. Second, the MATLAB results are validated using PowerWorld. Third, the transformer T1 tap ratio is changed to 0.99 and 1.01 to study its effect on the slack generator power. Fourth, the system is solved under a heavier loading condition. Finally, a three-phase short-circuit analysis is performed at Bus 3 and Bus 2 using $X_d'' = 0.2$ pu for all generators.

2. System Description

The studied system consists of five buses, two generators, two transformers, three transmission lines, and two load buses. Bus 1 is the slack bus, Bus 3 is the PV bus, and Buses 2, 4, and 5 are PQ buses. The system base is 100 MVA. The voltage base is 15 kV at Buses 1 and 3, while the voltage base is 345 kV at Buses 2, 4, and 5.

Table 1: Bus data used in the load flow study

Bus	Type	V (pu)	δ (deg)	P_G (pu)	Q_G (pu)	P_L (pu)	Q_L (pu)
1	Slack	1.00	0.00	–	–	0.00	0.00
2	PQ	–	–	0.00	0.00	4.00	1.20
3	PV	1.05	–	3.20	–	0.80	0.40
4	PQ	–	–	0.00	0.00	0.00	0.00
5	PQ	–	–	0.00	0.00	0.00	0.00

3. Theoretical Background and Derivation

3.1. Purpose of Power Flow Study

A power flow study determines the steady-state operating condition of a power system. The main unknowns are the voltage magnitude and voltage angle at the buses. After the bus voltages are known, the branch currents, branch power flows, generator powers, and system losses can be calculated.

Power flow studies are important because they help in planning, operation, and checking the performance of existing and future power networks. They also show whether bus voltages, branch loading, and generator powers are within acceptable limits.

3.2. Bus Admittance Matrix

The first step in the load flow solution is forming the bus admittance matrix, Y_{bus} . For a branch between bus i and bus k , the series impedance is:

$$z_{ik} = r_{ik} + jx_{ik} \quad (1)$$

The series admittance is:

$$y_{ik} = \frac{1}{z_{ik}} \quad (2)$$

For a normal line without a transformer tap, the elements added to Y_{bus} are:

$$Y_{ii} = Y_{ii} + y_{ik} + j\frac{B_{ik}}{2} \quad (3)$$

$$Y_{kk} = Y_{kk} + y_{ik} + j\frac{B_{ik}}{2} \quad (4)$$

$$Y_{ik} = Y_{ki} = -y_{ik} \quad (5)$$

For a transformer with off-nominal tap ratio a , the entries become:

$$Y_{ii} = Y_{ii} + \frac{y_{ik} + jB_{ik}/2}{a^2} \quad (6)$$

$$Y_{ik} = -\frac{y_{ik}}{a} \quad (7)$$

$$Y_{ki} = -\frac{y_{ik}}{a} \quad (8)$$

$$Y_{kk} = Y_{kk} + y_{ik} + j\frac{B_{ik}}{2} \quad (9)$$

Modeling note

The transformer tap appears directly in the Y_{bus} entries. This is why changing T1 from 1.00 to 0.99 or 1.01 changes the voltage profile and reactive power flow.

3.3. Voltage, Current, and Power Equations

The voltage at bus i in polar form is:

$$V_i = |V_i| \angle \delta_i \quad (10)$$

or:

$$V_i = |V_i|(\cos \delta_i + j \sin \delta_i) \quad (11)$$

The current injected into bus i is:

$$I_i = \sum_{k=1}^N Y_{ik} V_k \quad (12)$$

The complex power injected at bus i is:

$$S_i = P_i + jQ_i = V_i I_i^* \quad (13)$$

Substituting the current equation gives:

$$S_i = V_i \left(\sum_{k=1}^N Y_{ik} V_k \right)^* \quad (14)$$

Let:

$$Y_{ik} = G_{ik} + jB_{ik} \quad (15)$$

and:

$$\delta_{ik} = \delta_i - \delta_k \quad (16)$$

After expanding and separating the real and imaginary parts, the active and reactive power equations are:

$$P_i = \sum_{k=1}^N |V_i||V_k| (G_{ik} \cos \delta_{ik} + B_{ik} \sin \delta_{ik}) \quad (17)$$

$$Q_i = \sum_{k=1}^N |V_i||V_k| (G_{ik} \sin \delta_{ik} - B_{ik} \cos \delta_{ik}) \quad (18)$$

3.4. Scheduled Power and Mismatch Equations

At each bus, the scheduled active and reactive power injections are:

$$P_{i,\text{sch}} = P_{G_i} - P_{L_i} \quad (19)$$

$$Q_{i,\text{sch}} = Q_{G_i} - Q_{L_i} \quad (20)$$

The active and reactive power mismatches are:

$$\Delta P_i = P_{i,\text{sch}} - P_{i,\text{calc}} \quad (21)$$

$$\Delta Q_i = Q_{i,\text{sch}} - Q_{i,\text{calc}} \quad (22)$$

The load flow problem is solved when:

$$|\Delta P_i| < \epsilon \quad (23)$$

$$|\Delta Q_i| < \epsilon \quad (24)$$

3.5. Bus Classification

Each bus has four possible quantities: active power P , reactive power Q , voltage magnitude $|V|$, and voltage angle δ . Since only two equations are available at each bus, two quantities must be specified depending on the bus type.

Table 2: Bus types in power flow studies

Bus Type	Specified P	Specified Q	Specified $ V $	Specified δ
Slack Bus	No	No	Yes	Yes
PV Bus	Yes	No	Yes	No
PQ Bus	Yes	Yes	No	No

For this project, Bus 1 is the slack bus, Bus 3 is the PV bus, and Buses 2, 4, and 5 are PQ buses. The system has:

$$N = 5 \quad (25)$$

and one PV bus:

$$N_g = 1 \quad (26)$$

The total number of equations and unknowns is:

$$N_{\text{eq}} = 2N - N_g - 2 \quad (27)$$

$$N_{\text{eq}} = 2(5) - 1 - 2 = 7 \quad (28)$$

3.6. Newton–Raphson Method

The Newton–Raphson method is based on the first-order Taylor series expansion. For a nonlinear function $g(x)$:

$$g(x^{(k)} + \Delta x) \approx g(x^{(k)}) + \frac{\partial g}{\partial x} \Delta x \quad (29)$$

At the solution:

$$g(x^{(k)} + \Delta x) = 0 \quad (30)$$

Therefore:

$$0 = g(x^{(k)}) + J\Delta x \quad (31)$$

$$J\Delta x = -g(x^{(k)}) \quad (32)$$

In power flow form:

$$\begin{bmatrix} \Delta P \\ \Delta Q \end{bmatrix} = \begin{bmatrix} H & N \\ M & L \end{bmatrix} \begin{bmatrix} \Delta \delta \\ \Delta V \end{bmatrix} \quad (33)$$

where:

$$H = \frac{\partial P}{\partial \delta}, \quad N = \frac{\partial P}{\partial V}, \quad M = \frac{\partial Q}{\partial \delta}, \quad L = \frac{\partial Q}{\partial V} \quad (34)$$

The updates are:

$$\delta^{(k+1)} = \delta^{(k)} + \Delta \delta \quad (35)$$

$$V^{(k+1)} = V^{(k)} + \Delta V \quad (36)$$

3.7. Branch Currents and Losses

For a branch between bus i and bus k , the current from bus i to bus k is:

$$I_{ik} = \left(\frac{y_{ik} + jB_{ik}/2}{a^2} \right) V_i - \left(\frac{y_{ik}}{a} \right) V_k \quad (37)$$

The complex power flow from bus i to bus k is:

$$S_{ik} = V_i I_{ik}^* \quad (38)$$

The branch loss is:

$$S_{\text{loss},ik} = S_{ik} + S_{ki} \quad (39)$$

The total system losses are:

$$P_{\text{loss,total}} = \sum P_{\text{loss},ik} \quad (40)$$

$$Q_{\text{loss,total}} = \sum Q_{\text{loss},ik} \quad (41)$$

3.8. Short Circuit Formulation

For a three-phase fault, the prefault voltages are first obtained from the load flow solution. The generators are represented by their subtransient reactance:

$$X_d'' = 0.2 \text{ pu} \quad (42)$$

The short-circuit admittance matrix is formed and inverted:

$$Z_{\text{bus}} = Y_{\text{bus}}^{-1} \quad (43)$$

For a bolted three-phase fault at bus f :

$$Z_f = 0 \quad (44)$$

The fault current is:

$$I_f = \frac{V_f^0}{Z_{ff} + Z_f} \quad (45)$$

Since $Z_f = 0$:

$$I_f = \frac{V_f^0}{Z_{ff}} \quad (46)$$

The bus voltage during the fault is:

$$V_i^f = V_i^0 - Z_{if} I_f \quad (47)$$

4. Newton–Raphson Algorithm Flowchart

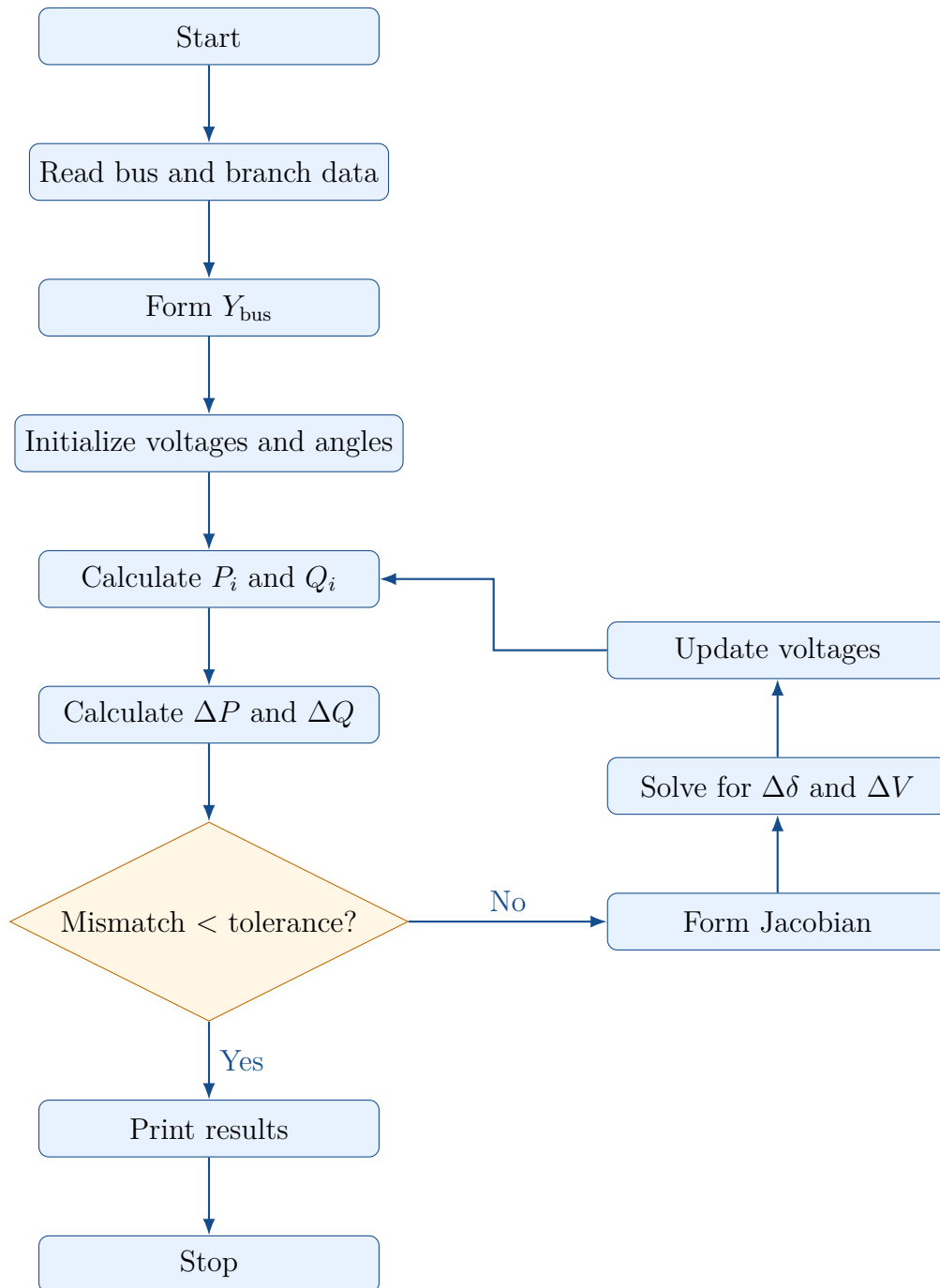


Figure 1: Newton–Raphson load flow algorithm

5. Task 1: Base Case Load Flow Results

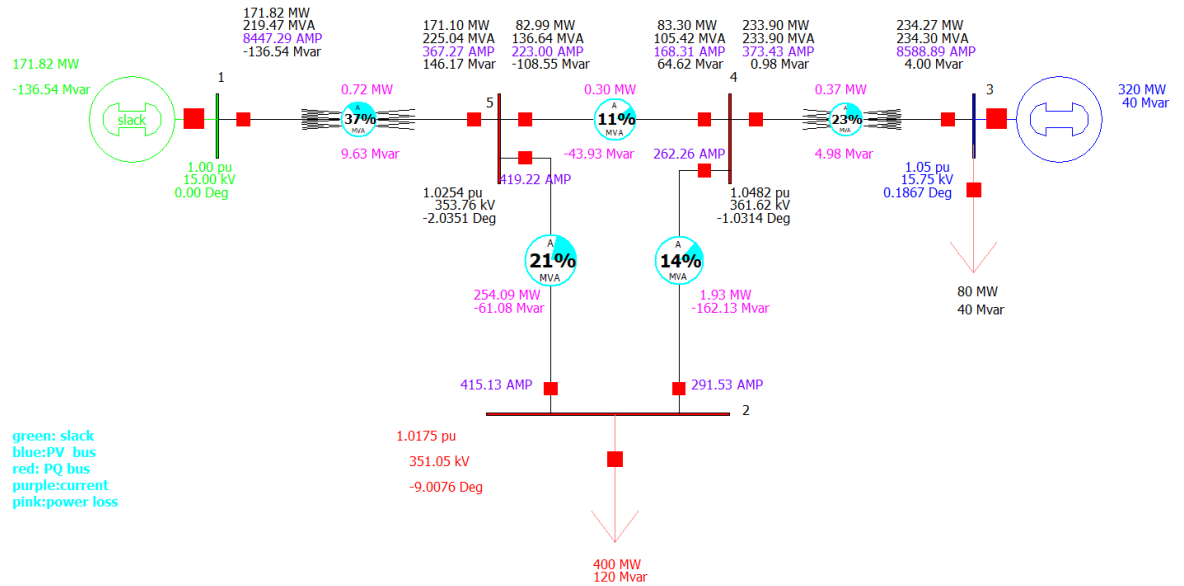


Figure 2: Base case load flow results in PowerWorld

Comment on Figure 2

The base-case PowerWorld diagram shows that the bus voltages are close to 1.0 pu, which indicates an acceptable voltage profile. Bus 1 acts as the slack bus and balances the system losses, while Bus 3 operates as the PV bus and maintains its voltage at 1.05 pu.

5.1. PowerWorld Bus Voltage Results

Table 3: Base case bus voltages from PowerWorld

Bus	Voltage (pu)	Voltage (kV)	Angle (deg)
1	1.0000	15.00	0.0000
2	1.0175	351.05	-9.0076
3	1.0500	15.75	0.1867
4	1.0482	361.62	-1.0314
5	1.0254	353.76	-2.0351

5.2. PowerWorld Slack and PV Bus Power

Table 4: Slack and PV generator power in the base case from PowerWorld

Bus	Bus Type	Active Power (MW)	Reactive Power (Mvar)
1	Slack	171.82	-136.54
3	PV	320.00	40.00

5.3. PowerWorld System Active Power Loss

$$P_{\text{loss}} = P_{\text{generation}} - P_{\text{load}} \quad (48)$$

$$P_{\text{loss}} = (171.82 + 320.00) - (400.00 + 80.00) \quad (49)$$

$$P_{\text{loss}} = 11.82 \text{ MW} \quad (50)$$

Base-case PowerWorld active power balance

$$P_{\text{generation}} = 491.82 \text{ MW}$$

$$P_{\text{load}} = 480.00 \text{ MW}$$

$$P_{\text{loss}} = 11.82 \text{ MW}$$

5.4. MATLAB Base Case Results

NR load flow converged in 5 iterations.

A) VOLTAGE AND ANGLE AT EACH BUS

Bus	Voltage_pu	Angle_deg
1	1.0000	0
2	1.0175	-8.9213
3	1.0500	0.34674
4	1.0482	-0.90129
5	1.0254	-1.9709

B) SLACK BUS POWER

Slack Bus = 1
P_slack = 1.661 pu = 166.10 MW
Q_slack = -1.3632 pu = -136.32 Mvar

C) PV BUS POWER

PV Bus = 3
P_PV = 3.2 pu = 320.00 MW
Q_PV = 0.43854 pu = 43.854 Mvar

Listing 1: MATLAB base-case load flow output

Comparison with PowerWorld

, which shows that the Newton–Raphson algorithm reached a stable solution. The bus voltages are very close to the PowerWorld values. The slack active power from MATLAB is 166.10 MW, while PowerWorld shows 171.82 MW. This difference can happen because of small differences in line charging, transformer modeling, tap representation, or entered branch parameters. The reactive power at the slack bus is very close in both tools.

5.5. MATLAB Branch Current Results

D) CURRENT FLOW IN EACH BRANCH							
Branch		I_from_pu	I_from_kA	I_to_pu	I_to_kA	P_from_MW	Q_from_Mvar
	P_to_MW Q_to_Mvar						
Line 1: Bus 2 to Bus 4		1.7476	0.29245	1.5741	0.26342	-149.46	-96.34
	151.41 -65.558						
Line 2: Bus 2 to Bus 5		2.4732	0.41389	2.4973	0.41792	-250.54	-23.66
	253.29 -37.611						
Line 3: Bus 4 to Bus 5		1.0407	0.17416	1.3575	0.22717	88.203	64.186
	-87.883 -107.94						
T1: Bus 1 to Bus 5		2.1488	8.2706	2.1488	0.35959	166.10	-136.32
	-165.41 145.55						
T2: Bus 3 to Bus 4		2.2860	8.7989	2.2860	0.38256	240.00	3.8537
	-239.61 1.3722						

Listing 2: MATLAB branch current output for base case

Interpretation of Branch Current Results

generator transformer sides because the current base at 15 kV is much higher than the current base at 345 kV. The branch power values also show that power flows from the generator buses toward the load buses, while the small differences between sending-end and receiving-end active powers represent branch losses.

5.6. MATLAB System Losses

From the MATLAB results:

$$P_{\text{loss, MATLAB}} = (166.10 + 320.00) - (400.00 + 80.00) \quad (51)$$

$$P_{\text{loss, MATLAB}} = 6.10 \text{ MW} \quad (52)$$

MATLAB active power loss

The MATLAB active power loss is approximately:

$$P_{\text{loss, MATLAB}} = 6.10 \text{ MW}$$

6. Task 2: Validation Using PowerWorld

The MATLAB results were validated using PowerWorld Simulator. The comparison focused on bus voltage magnitudes, bus voltage angles, slack bus power, PV bus power, branch currents, and system losses.

Table 5: MATLAB and PowerWorld base-case comparison

Bus	V_{MATLAB} (pu)	V_{PW} (pu)	δ_{MATLAB} (deg)	δ_{PW} (deg)
1	1.0000	1.0000	0.0000	0.0000
2	1.0175	1.0175	-8.9213	-9.0076
3	1.0500	1.0500	0.3467	0.1867
4	1.0482	1.0482	-0.9013	-1.0314
5	1.0254	1.0254	-1.9709	-2.0351

Table 6: Main generator power comparison

Quantity	MATLAB	PowerWorld	Difference	Comment
Slack P_G	166.10 MW	171.82 MW	5.72 MW	Different loss/modeling details
Slack Q_G	-136.32 Mvar	-136.54 Mvar	0.22 Mvar	Very close
PV P_G	320.00 MW	320.00 MW	0.00 MW	Matched
PV Q_G	43.854 Mvar	40.00 Mvar	3.854 Mvar	Close

7. Task 3: Effect of Transformer T1 Tap Change

In this task, the load flow was solved twice after changing transformer T1 tap ratio. The two studied tap values are 0.99 and 1.01. The objective is to observe how changing the transformer tap affects the active and reactive power generated by the slack bus.

7.1. T1 Tap = 0.99

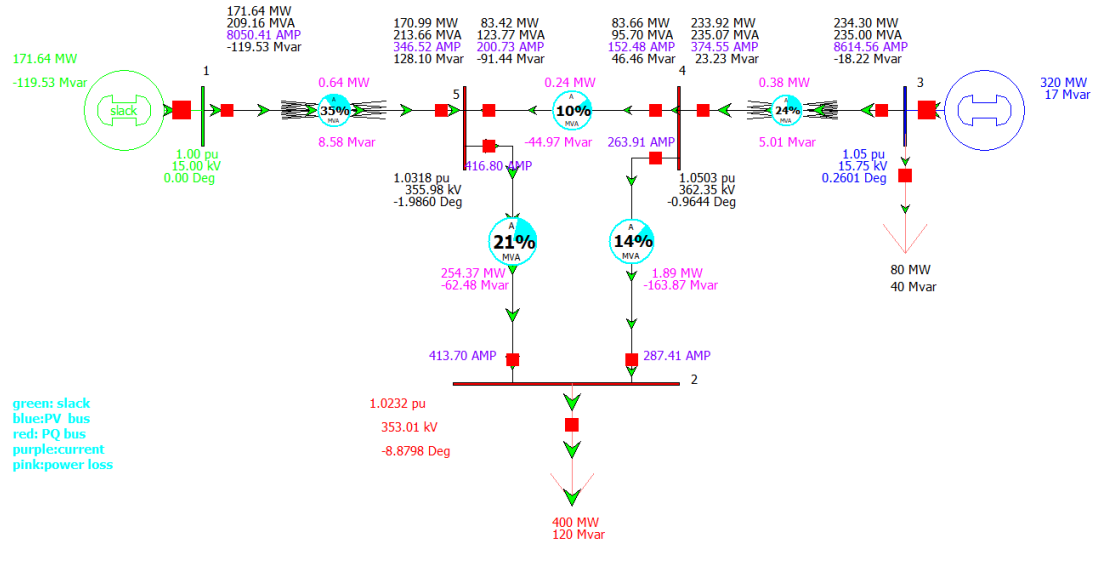


Figure 3: PowerWorld load flow results for T1 tap = 0.99

Comment on Figure 3

When the T1 tap is set to 0.99, the slack bus reactive power becomes less negative. This means the slack bus absorbs less reactive power compared with the base case. The voltage profile also changes slightly around Buses 5 and 2.

7.2. T1 Tap = 1.01

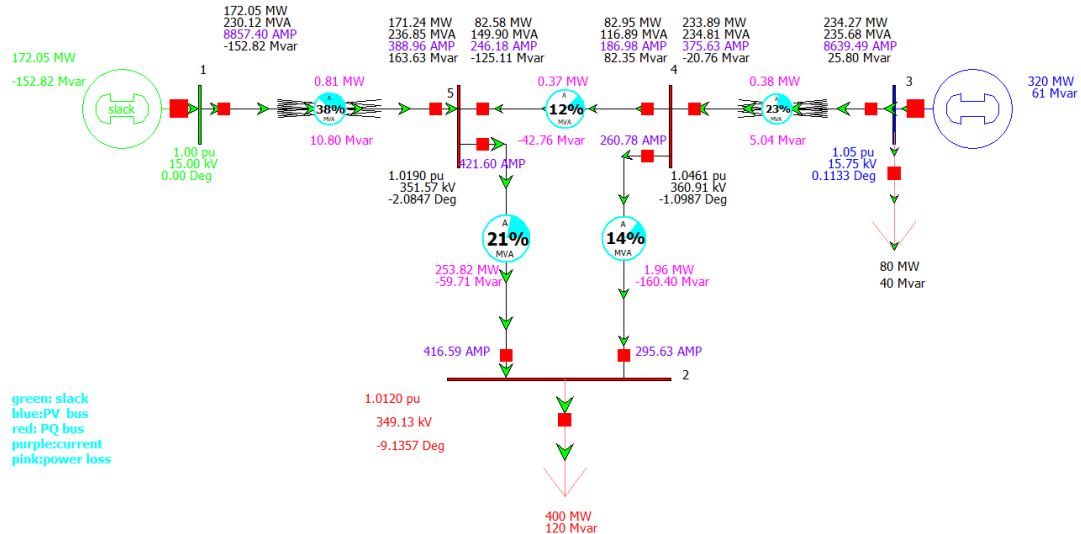


Figure 4: PowerWorld load flow results for T1 tap = 1.01

Comment on Figure 4

When the T1 tap is set to 1.01, the slack bus reactive power becomes more negative. This means the slack bus absorbs more reactive power compared with the base case. The active power changes only slightly, while the reactive power changes more clearly.

7.3. MATLAB T1 Tap Study Results

SUMMARY: EFFECT OF T1 TAP ON SLACK BUS POWER

Tap	Slack_P_pu	Slack_Q_pu	Slack_P_MW	Slack_Q_Mvar
0.99	1.659	-1.193	165.90	-119.30
1.01	1.6632	-1.5263	166.32	-152.63

Listing 3: MATLAB summary for T1 tap study

Table 7: MATLAB effect of T1 tap ratio on slack bus power

Case	T1 Tap	Slack P_G (MW)	Slack Q_G (Mvar)	Observation
Base case	1.00	166.10	-136.32	Reference case
Case 1	0.99	165.90	-119.30	Reactive absorption decreases
Case 2	1.01	166.32	-152.63	Reactive absorption increases

Table 8: MATLAB change in slack bus power compared with base case

T1 Tap	ΔP_G (MW)	ΔQ_G (Mvar)	Comment
0.99	-0.20	+17.02	Slack bus absorbs less reactive power
1.01	+0.22	-16.31	Slack bus absorbs more reactive power

Comment on MATLAB T1 tap results

The MATLAB results show the same trend as the PowerWorld results. The active power generated by the slack bus changes by less than 1 MW, while the reactive power changes by about 16–17 Mvar. This confirms that transformer tap changes mainly influence voltage and reactive power distribution.

7.4. PowerWorld T1 Tap Study Results**Table 9:** PowerWorld effect of T1 tap ratio on slack bus power

Case	T1 Tap	Slack P_G (MW)	Slack Q_G (Mvar)	Observation
Base case	1.00	171.82	-136.54	Reference case
Case 1	0.99	171.64	-119.53	Reactive absorption decreases
Case 2	1.01	172.05	-152.82	Reactive absorption increases

Summary of Tap Change Effect

has a small effect on slack active power and a stronger effect on slack reactive power. This agrees with the physical role of transformer taps, because tap changes mainly affect voltage magnitudes and reactive power flow.

8. Task 4: Load Flow Under New Loading Conditions

In this task, the system was solved under a heavier loading condition using PowerWorld. The main loading change is at Bus 2, where the load increases to 800 MW and 280 Mvar. The purpose is to evaluate the voltage profile and system performance under stressed loading.

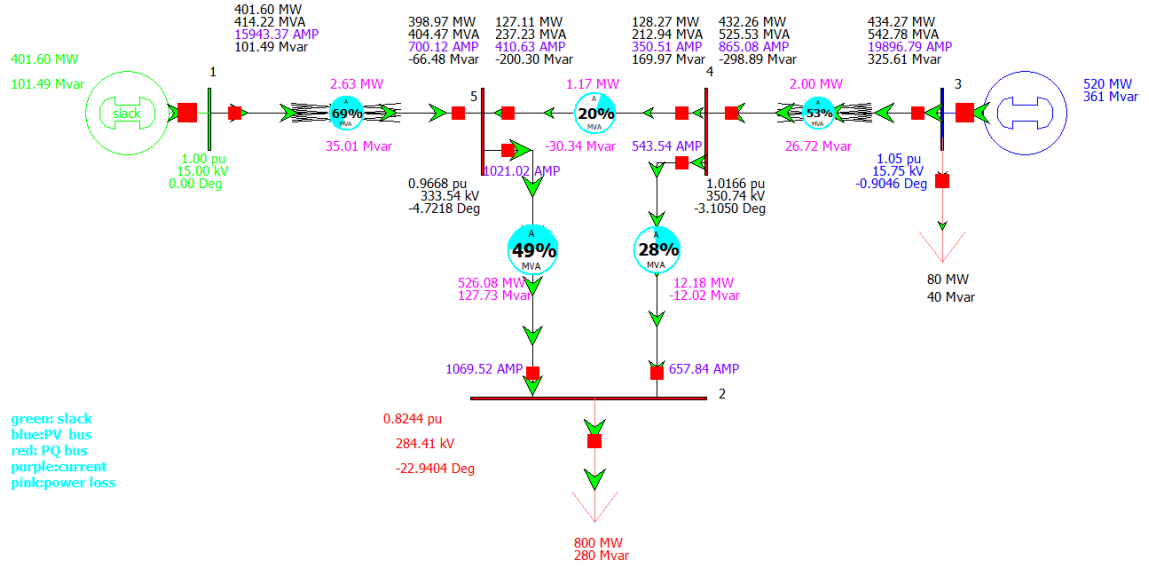


Figure 5: PowerWorld results under the new loading condition

Comment on Figure 5

The heavy loading case causes a large voltage drop at Bus 2. The Bus 2 voltage decreases to 0.8244 pu, which indicates a stressed operating condition. The slack generator and PV generator both supply higher power, and the system losses increase significantly.

8.1. Main Results Under New Loading Condition

Table 10: Main results for the new loading condition

Quantity	Base Case	New Loading Case	Change
Bus 2 load	400 MW, 120 Mvar	800 MW, 280 Mvar	Increased
Slack bus power	171.82 MW, -136.54 Mvar	401.60 MW, 101.49 Mvar	Increased
PV bus power	320 MW, 40 Mvar	520 MW, 361 Mvar	Increased
Bus 2 voltage	1.0175 pu	0.8244 pu	Large voltage drop
Bus 5 voltage	1.0254 pu	0.9668 pu	Voltage decreased

8.2. Active Power Loss Under New Loading Condition

$$P_{\text{loss}} = P_{\text{generation}} - P_{\text{load}} \quad (53)$$

$$P_{\text{loss}} = (401.60 + 520.00) - (800.00 + 80.00) \quad (54)$$

$$P_{\text{loss}} = 41.60 \text{ MW} \quad (55)$$

Heavy-loading active power balance

$$P_{\text{generation}} = 921.60 \text{ MW}$$

$$P_{\text{load}} = 880.00 \text{ MW}$$

$$P_{\text{loss}} = 41.60 \text{ MW}$$

Effect of Heavy Loading on System Losses

in the base case to 41.60 MW in the heavy loading case. This is expected because higher loading increases branch currents, and resistive losses increase approximately with I^2R .

9. Task 5: Three-Phase Short Circuit Analysis

For the short-circuit study, a balanced three-phase fault is applied at Bus 3 and Bus 2. The generator subtransient reactance is taken as:

$$X_d'' = 0.2 \text{ pu} \quad (56)$$

The three-phase fault current is calculated using the Thevenin equivalent impedance seen from the faulted bus:

$$I_f = \frac{V_{\text{prefault}}}{Z_{\text{th}} + Z_f} \quad (57)$$

For a bolted three-phase fault:

$$Z_f = 0 \quad (58)$$

Therefore:

$$I_f = \frac{V_{\text{prefault}}}{Z_{\text{th}}} \quad (59)$$

9.1. MATLAB Pre-Fault Voltages

Pre-fault NR load flow converged in 5 iterations.

Bus	PreFault_V_pu	PreFault_Angle_deg
1	1.0000	0
2	1.0175	-8.9213
3	1.0500	0.34674
4	1.0482	-0.90129
5	1.0254	-1.9709

Listing 4: MATLAB pre-fault bus voltages before short-circuit analysis

Comment on pre-fault voltages

The pre-fault voltages are the same as the solved load-flow voltages. These voltages are used as the initial operating condition before applying the three-phase fault.

9.2. Fault at Bus 3

```
TASK 5 RESULTS: 3-PHASE FAULT AT BUS 3

Total fault current at Bus 3:
|If| = 9.425239 pu
Angle = -89.2315 degrees
|If| = 36.277760 kA

Bus results during fault:
Bus 1: V = 0.16618 pu, generator contribution = 3.3145 pu = 12.757 kA
Bus 2: V = 0.17957 pu
Bus 3: V = 0 pu, generator contribution = 6.4351 pu = 24.769 kA
Bus 4: V = 0.047076 pu
Bus 5: V = 0.11735 pu
```

Listing 5: MATLAB short-circuit results for fault at Bus 3

Comment on Bus 3 fault

The Bus 3 voltage becomes approximately zero because the fault is applied directly at Bus 3. The total fault current is high because Bus 3 is directly connected to a generator. The generator at Bus 3 gives the largest contribution to the fault current.

9.3. Fault at Bus 2

```
TASK 5 RESULTS: 3-PHASE FAULT AT BUS 2

Total fault current at Bus 2:
|If| = 7.192481 pu
Angle = -97.4568 degrees
|If| = 1.203647 kA

Bus results during fault:
Bus 1: V = 0.30570 pu, generator contribution = 2.4876 pu = 9.5747 kA
Bus 2: V = 0 pu, total fault current = 7.1925 pu = 1.2036 kA
Bus 3: V = 0.34716 pu, generator contribution = 4.7493 pu = 18.28 kA
Bus 4: V = 0.30684 pu
Bus 5: V = 0.25624 pu
```

Listing 6: MATLAB short-circuit results for fault at Bus 2

Comment on Bus 2 fault

The Bus 2 voltage becomes zero because the fault is applied directly at Bus 2. The fault current in pu is lower than the Bus 3 fault current. The current in kA is much smaller because Bus 2 is a 345 kV bus, so its current base is lower than the 15 kV generator bus current base.

9.4. Fault Current Comparison

Table 11: Three-phase fault current comparison

Fault Location	Fault Current (pu)	Fault Current Angle (deg)	Fault Current (kA)
Bus 3	9.4252	-89.231	36.278
Bus 2	7.1925	-97.457	1.2036

Difference between Bus 3 and Bus 2 fault currents

$$|I_{SC,3}| - |I_{SC,2}| = 2.232758 \text{ pu}$$

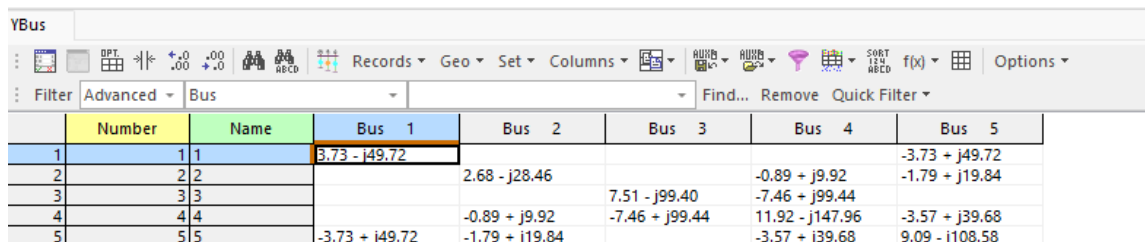
$$|I_{SC,3}| - |I_{SC,2}| = 35.074114 \text{ kA}$$

Final comment on Task 5

The fault current at Bus 3 is larger than the fault current at Bus 2. This is expected because Bus 3 is directly connected to a generator and has a smaller equivalent impedance seen from the fault location. Bus 2 is electrically farther from the generators through the transmission network, so the equivalent impedance is higher. Also, the kA comparison is affected by the different voltage bases: Bus 3 is on the 15 kV side, while Bus 2 is on the 345 kV side.

10. PowerWorld Matrices

10.1. Y-Bus Matrix



	Number	Name	Bus 1	Bus 2	Bus 3	Bus 4	Bus 5
1	1	1	3.73 - j49.72				-3.73 + j49.72
2	2	2		2.68 - j28.46		-0.89 + j9.92	-1.79 + j19.84
3	3	3			7.51 - j99.40	-7.46 + j99.44	
4	4	4		-0.89 + j9.92	-7.46 + j99.44	11.92 - j147.96	-3.57 + j39.68
5	5	5	-3.73 + j49.72	-1.79 + j19.84		-3.57 + j39.68	9.09 - j108.58

Figure 6: Y-Bus matrix from PowerWorld

10.2. Jacobian Matrix

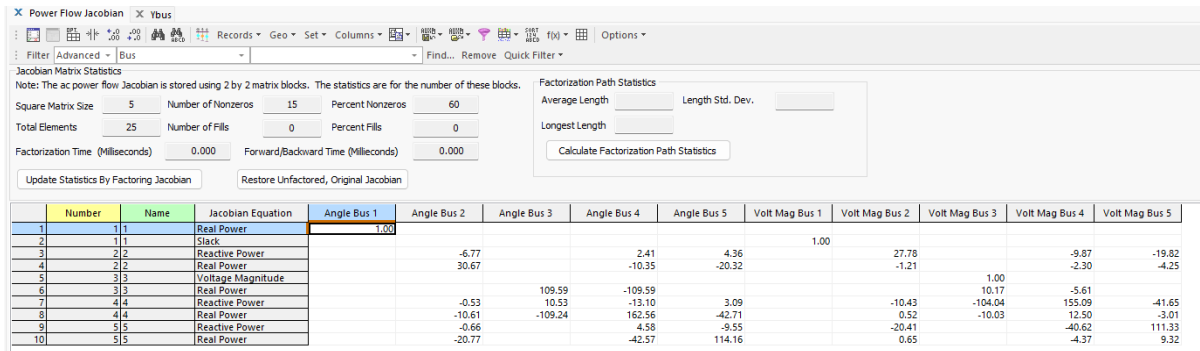


Figure 7: Jacobian matrix for the base-case operating condition

Role of the Jacobian Matrix

to the voltage corrections in the Newton–Raphson method. A well-conditioned Jacobian helps the solution converge quickly.

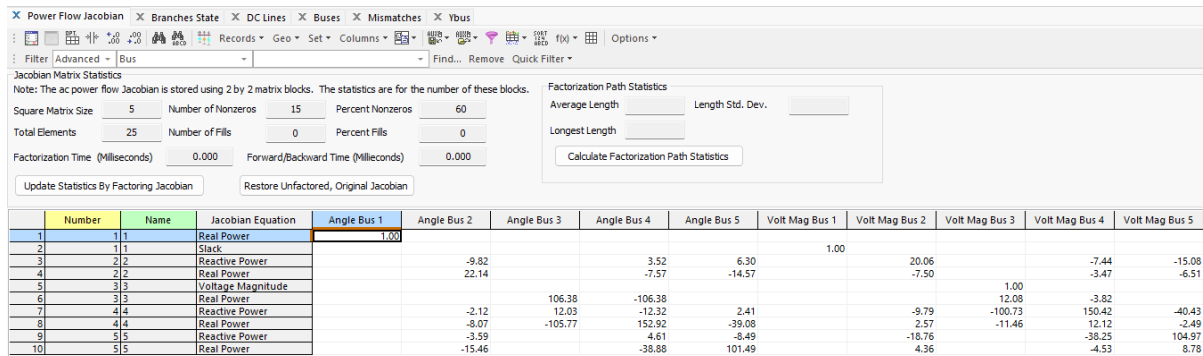


Figure 8: Jacobian matrix for the heavy-loading operating condition

Jacobian Update at Each Iteration

, voltage angles, and therefore the numerical values inside the Jacobian matrix. This is why the Jacobian is updated at every Newton–Raphson iteration.

11. Conclusion

The five-bus power system was studied using MATLAB and PowerWorld. In the base case, the voltage profile was acceptable, and the system operated close to nominal voltage levels. The MATLAB Newton–Raphson code converged in five iterations and produced voltage results close to PowerWorld. The tap-changing study showed that transformer T1 mainly affects the slack bus reactive power rather than the active power. The heavier loading case caused a large voltage drop at Bus 2 and increased the active power losses.

Finally, the short-circuit study showed that the fault at Bus 3 produces a larger current than the fault at Bus 2 because Bus 3 is electrically closer to a generator.

References

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